RESEARCH OBJECTIVE 1 · DATASET PAPER

Construction of a Sector-Wise Corporate Event and Price Shock Dataset for Nifty 50 Stocks

NSE India · 2018 – 2024 · 50 Companies · 11 Sectors

1. Research Overview

This objective proposes the construction of **NiftyEvent-50**, the first publicly available, sector-wise labeled dataset mapping NSE corporate events to quantified price shock outcomes. Existing event-driven datasets (EDT, FNSPID) cover only US markets. No equivalent labeled benchmark exists for Indian equities, leaving portfolio risk models trained on NSE data without a reliable evaluation corpus. This paper fills that gap with a reproducible scraping, alignment, and labeling pipeline covering all Nifty 50 stocks across 11 sectors for the period 2018–2024.

Dataset Paper	NSE India	2018–2024	50 Stocks	11 Sectors	5 Event Types

2. Research Gap

- All major event-driven financial datasets (EDT, FNSPID, FinBen) are built on US exchanges (NYSE / NASDAQ) — none cover NSE.
- Researchers working on Indian markets currently scrape headlines manually with no standardised, reproducible pipeline.
- No dataset labels corporate events alongside both price shock magnitude AND volatility spike together for Indian stocks.
- Portfolio risk models for emerging markets lack a crisis-free baseline event benchmark to train classifiers on.

3. Objective Statement

To design and release a reproducible data collection and labeling pipeline that captures five corporate event types for all Nifty 50 companies, aligns each event to a ±8-day OHLCV window, and produces a structured dataset with price shock percentage, volatility spike flag, and directional label — suitable for training and benchmarking portfolio risk classifiers.

4. Step-by-Step Implementation Flow

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Step 1: Define Event Taxonomy

- Select 5 corporate event types: Dividend Declaration, Bonus Issue, Stock Split, Earnings Surprise (Beat/Miss), Management Change.
- Define precise detection rules per event type (e.g., earnings surprise = actual EPS vs analyst consensus deviation > 5%).
- Create a master event catalog template with required metadata fields.

Tools: *NSE Corporate Announcements Portal, BSE XML feed*

2 Step 2: Collect Event Announcements

• Scrape NSE Corporate Announcements page for all 50 Nifty tickers (2018–2024).

• Parse announcement date, ticker, event type, and brief description.

• Filter and deduplicate: keep only the 5 defined event types; remove duplicates within ±2 days of same event-ticker pair.

• Store raw events in structured CSV: ~3,000–5,000 raw records expected.

Tools: *Python, BeautifulSoup / Scrapy, pandas*

3 Step 3: Fetch OHLCV Price Windows

• For each event date, fetch OHLCV data using yfinance: 3 trading days before and 5 trading days after.

• Compute pre-event baseline price: average Close of the 3 days before.

• Compute post-event price: average Close of the 5 days after.

• Handle missing days (holidays, circuit breakers) using forward-fill within the window.

Tools: *yfinance, pandas, numpy*

4 Step 4: Compute Price Shock and Volatility Labels

• Price Shock %: ((post_price - pre_price) / pre_price) x 100.

• Direction Label: Up if shock > +0.5%, Down if shock < -0.5%, Neutral otherwise.

• Volatility Spike: compute rolling 30-day std dev of daily returns; flag as Yes if window std dev > 1.5x the rolling average.

• Store all computed values alongside raw prices in final dataset.

Tools: *pandas, numpy*

5 Step 5: Sector Mapping and Quality Checks

• Map each ticker to its NSE sector classification (11 sectors: IT, Banking, Energy, FMCG, Auto, Pharma, Metal, Realty, Telecom, Finance, Infra).

• Run completeness check: flag any row with > 2 missing values.

• Run spot-check on 100 randomly sampled events: manually verify event date, price data, and label correctness.

• Target: >= 90% labeling accuracy on spot-check sample.

Tools: *pandas, Excel for spot-check*

6 Step 6: Release and Documentation

• Export final dataset as CSV and Excel (.xlsx) with a data dictionary.

• Upload to Zenodo or Hugging Face Datasets with DOI for citability.

• Write a short data descriptor paper: pipeline description, dataset statistics, coverage analysis, and baseline classifier results.

• Release scraping code as open-source GitHub repository.

Tools: *Zenodo / Hugging Face, GitHub, Markdown*

5. Output Dataset Schema

Column	Type	Description / Example
date	DATE	2021-07-15
ticker	TEXT	RELIANCE.NS
sector	TEXT	Energy
event_type	TEXT	Dividend Declaration

Column	Type	Description / Example
pre_event_price	FLOAT	2105.30 (3-day avg before)
post_event_price	FLOAT	2198.75 (5-day avg after)
shock_pct	FLOAT	+4.44 (price change %)
volatility_label	BOOL	Yes / No (spike > 1.5x avg)
direction_label	TEXT	Up / Down / Neutral

6. Success / Evaluation Metrics

Metric	Target Value
Total headlines / events collected	>= 3,000 labeled events
Stocks covered	All 50 Nifty 50 companies
Sectors covered	11 NSE sectors
Event types	5 (dividend, bonus, split, earnings, mgmt)
Time span	2018 – 2024 (6 years)
Labeling accuracy (manual spot-check)	>= 90%
Missing data tolerance	< 2% per column

7. Tools, Libraries & Timeline

Task	Tool / Library	Week
Event scraping & deduplication	Python, Scrapy, BeautifulSoup	1–2
OHLCV fetch & window alignment	yfinance, pandas	2–3
Label computation (shock, volatility)	numpy, pandas	3
Sector mapping & quality checks	pandas, Excel	4
Baseline classifier (optional)	scikit-learn (Random Forest)	5
Documentation & release	Zenodo, GitHub, Markdown	6

8. Target Conference Venues

[Primary] IEEE ICDCS 2025 — Data & Benchmark Track

[Primary] ACM ICAIF 2025 — Dataset Paper Track

[Secondary] Springer ICMLSC — Emerging Markets Data

[Secondary] ICACCS 2025 — Indian Market Analysis

9. Files to be Submitted

All files must be submitted as a single zipped folder named **StudentName_Objective1/** containing the following:

Category	File Name	Description
Raw Data	**raw_events.csv**	Scraped NSE announcements: ticker, date, event type, description
Raw Data	**ohlcv_windows/**	Folder of per-ticker OHLCV CSVs (3 days before, 5 days after each event)
Final Dataset	**NiftyEvent50_final.csv**	Complete labeled dataset with all 9 schema columns
Final Dataset	**data_dictionary.xlsx**	Column definitions, label rules, and threshold values used
Validation	**spot_check_100.xlsx**	100 manually verified rows with correct/incorrect flags and accuracy score
Validation	**dataset_statistics.ipynb**	Coverage report: events per sector, per event type, per year, missing data %
Code	**scraper.py**	NSE announcement scraping script
Code	**ohlcv_fetch.py**	yfinance price window fetching script
Code	**label_compute.py**	Shock %, volatility flag, and direction label computation
Code	**requirements.txt**	Python library names and versions used

10. How the Supervisor Uses These Files

File	Used For
NiftyEvent50_final.csv	Feed into parent system backtesting node (O3) as event-based signal evaluation benchmark
data_dictionary.xlsx	Cite in methodology section of research paper for data provenance
spot_check_100.xlsx	Report labeling accuracy figure in dataset paper
dataset_statistics.ipynb	Extract coverage statistics and charts for paper figures
requirements.txt	Include in reproducibility appendix of the paper